Page 13 Exercises

Let A denote associativity of integers, C denote commutativity of integers. Here, a, b, c, d, x, w, y, z are all integers.

1 LHS $= (a+b)+(c+d)$

$= a+b+c+d$ [A]

$= a+d+b+c$ [C]

$= (a+d)+(b+c)$ [A]

$=$ RHS

2 LHS $= (a+b)+(c+d)$

$= a+b+c+d$ [A]

$= a+c+b+d$ [C]

$= (a+c)+(b+d)$ [A]

$=$ RHS

3 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a+c-b-d$ [C]

$= (a+c)+(-b-d)$ [A]

$=$ RHS

4 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a+c-b-d$ [C]

$= (a+c)+(-b-d)$ [A]

$= (a+c)-(b+d)$ [N5]

$=$ RHS

5 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= a-d+c-b$ [C]

$= (a-d)+(c-b)$ [A]

$=$ RHS

6 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= -b-d+a+c$ [C]

$= (-b-d)+(a+c)$ [A]

$= -(b+d)+(a+c)$ [N5]

$=$ RHS

7 LHS $= (a-b)+(c-d)$

$= a-b+c-d$ [A]

$= -b-d+a+c$ [C]

$= -b-d-(-a)-(-c)$ [N4]

$= (-b-d)+(-(-a)-(-c))$ [A]

$= -(b+d)-(-a-c)$ [N5]

$=$ RHS

8 LHS $= ((x+y)+z)+w$

$= x+y+z+w$ [A]

$= x+z+y+w$ [C]

$= (x+z)+(y+w)$ [A]

$=$ RHS

9 LHS $= (x-y)-(z-\omega)$

$= x-y-z-(-\omega)$ [A]

$= x-y-z+\omega$ [N4]

$= x+\omega-y-z$ [C]

$= (x+\omega)-y-z$ [A]

$=$ RHS

10 $(x-y)-(z-\omega)$

$= x-y-z-(-\omega)$ [A]

$= x-y-z+\omega$ [N4]

$= x-z+\omega-y$ [C]

$= (x-z)+(\omega-y)$ [A]

$=$ RHS

11 LHS $= -(\alpha+b+c)$

$= -\alpha-b-c$ [N5]

$= -\alpha+(-b)+(-c)$ [A]

$=$ RHS

12 LHS $= -(\alpha-b-c)$

$= -(\alpha)-(-b)-(-c)$ [N5]

$= -\alpha+b+c$ [N4]

$=$ RHS

13 LHS $= -(\alpha-b)$

$= -(-b+\alpha)$ [C]

~~$= -(-b)-\alpha$ [A]~~ $-(-b)-\alpha$ [A]

$= b-\alpha$ [N4]

$=$ RHS

14 $-2+x=4$

$x = 4+2 = 6$

15 $2-x=5$

$x = 2-5 = -3$

16 $x-3=7$

$x = 10$

17 $-x+4=-1$

$x=5$

18 $4-x=8$

$x=-4$

19 $-5-x=-2$

$x=-3$

20 $-7+x=-10$

$x=-3$

21 $-3+x=4$

$x=7$

22 $a+b=\alpha+c$

$\alpha+b-\alpha=\alpha+c-\alpha$ [subtracting α from both]

$b=c$

$\therefore a+b=\alpha+c \iff b=c$

23 $a+b=\alpha$

$a+b-\alpha=\alpha-\alpha$ [subtracting α from both]

$b=0$

$\therefore \alpha+b=\alpha \iff b=0$ (N1 is proved)

Page 21 Exercises

1 a) $8a^2b^3(27a^4)(2^5ab)$

$= 2^8 3^3 a^7 b^4$

b) $16b^3a^2(6ab^4)(ab)^3$

$= 2^5 3^1 a^6 b^8$

c) $3^2(2ab)^3(16a^2b^5)(24b^2a)$

$= 2^{10} 3^3 a^6 b^{10}$

d) $24a^3(2ab^2)^3(3ab)^2$

$= 2^6 3^3 a^8 b^8$

e) $(3ab)^2(27a^3b)(16ab^5)$

$= 2^4 3^6 a^6 b^8$

f) $32a^4b^5a^3b^2(6ab^3)^4$

$= 2^9 3^4 a^{11} b^{19}$

2 $(a+b)^2(a+b)$

$= (a^2+2ab+b^2)(a+b)$

$= a^3+a^2b+2a^2b+2ab^2+b^2a+b^3$

$= a^3+3a^2b+3ab^2+b^3$

$(a-b)^2(a-b)$

$= (a^2-2ab+b^2)(a-b)$

$= a^3-a^2b-2a^2b+2ab^2+b^2a-b^3$

$= a^3-3a^2b+3ab^2-b^3$

3 $(a+b)^2(a+b)^2$

$= (a^2+2ab+b^2)(a^2+2ab+b^2)$

$= a^4+2a^3b+b^2a^2+2a^3b+4a^2b^2+2ab^3+b^2a^2+2ab^3+b^4$

$= a^4+4a^3b+6a^2b^2+4ab^3+b^4$

$(a-b)^2(a-b)^2$

$= (a^2-2ab+b^2)(a^2-2ab+b^2)$

$= a^4-2a^3b+a^2b^2-2a^3b+4a^2b^2-2ab^3+b^2a^2-2ab^3+b^4$

$= a^4-4a^3b+6a^2b^2-4ab^3+b^4$

4 $(2-4x)^2 = 4 - 16x + 16x^2$

5 $(1-2x)^2 = 1 - 4x + 4x^2$

6 $(2x+5)^2 = 4x^2 + 20x + 25$

7 $(x-1)^2 = x^2 - 2x + 1$

8 $(x+1)(x-1) = x^2 - x + x - 1 = x^2 - 1$

9 $(2x+1)(x+5) = 2x^2 + 10x + x + 5 = 2x^2 + 11x + 5$

10 $(x^2+1)(x^2-1) = x^4 - x^2 + x^2 - 1 = x^4 - 1$

11 $(1+x^3)(1-x^3) = 1 - x^6$

12 $(x^2+1)^2 = x^4 + 2x^2 + 1$

13 $(x^2-1)^2 = x^4 - 2x^2 + 1$

14 $(x^2+2)^2 = x^4 + 4x^2 + 4$

15 $(x^2-2)^2 = x^4 - 4x^2 + 4$

16 $(x^3-4)^2 = x^6 - 8x^3 + 16$

17 $(x^3-4)(x^3+4) = x^6 - 16$

18 $(2x^2+1)(2x^2-1) = 4x^4 - 1$

19 $(-2+3x)(-2-3x) = 4 - 9x^2$

20 $(x+1)(2x+5)(x-2) = (x^2-2x+x-2)(2x+5) = (x^2-x-2)(2x+5)$

~~$= 2x^3 + 5x^2 - 4x^2 - 10x$~~

$= 2x^3 + 5x^2 - 2x^2 - 5x - 4x - 10$

$= 2x^3 + 3x^2 - 9x - 10$

21 $(2x+1)(1-x)(3x+2)$

$= (6x^2 + 4x + 3x + 2)(1-x)$

$= (6x^2 + 7x + 2)(1-x)$

$= 6x^2 - 6x^3 + 7x - 7x^2 + 2 - 2x$

$= -[6x^3 + x^2 - 5x - 2]$

22 $(3x-1)(2x+1)(x+4)$

$= (6x^2 + 3x - 2x - 1)(x+4)$

$= (6x^2 + x - 1)(x+4)$

$= 6x^3 + 24x^2 + x^2 + 4x - x - 4$

$= 6x^3 + 25x^2 + 3x - 4$

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23 $(-1-x)(-2+x)(1-2x)$

$= (2-x+2x-x^2)(1-2x)$

$= (2+x-x^2)(1-2x)$

$= 2-4x+x-2x^2-x^2+2x^3$

$= 2x^3-3x^2-3x+2$

24 $(-4x+1)(2-x)(3+x)$

$= (6+2x-3x-x^2)(-4x+1)$

$= (6-x-x^2)(-4x+1)$

$= -24x+6-4x^2-x+4x^3-x^2$

$= 4x^3-5x^2-25x+6$

25 $(1-x)(1+x)(2-x)$

$= (1-x^2)(2-x)$

$= 2-x-2x^2+x^3$

$= x^3-2x^2-x+2$

26 $(x-1)^2(3-x)$

$= (x^2-2x+1)(3-x)$

$= 3x^2-x^3-6x+2x^2+3-x$

$= -[x^3-5x^2+7x-3]$

27 $(1-x)^2(2-x)$

$= (1-2x+x^2)(2-x)$

$= 2-x-4x+2x^2+2x^2-x^3$

$= -[x^3-4x^2+5x-2]$

28 $(1-2x)^2(3+4x)$

$= (1-4x+4x^2)(3+4x)$

$= 3+4x-12x-16x^2+12x^2+16x^3$

$= 16x^3-4x^2-8x+3$

29 $(2x+1)^2(2-3x)$

$= (4x^2+4x+1)(2-3x)$

$= 8x^2-12x^3+4x-12x^2+2-3x$

$= -[12x^3+4x^2-x-2]$

30 Let population be denoted by P.

According to given conditions

$$P_{(t)} = 50{,}000 \times 2^{t/10}$$

(a) 1970 $\Rightarrow t = 60 \Rightarrow t/10 = 6 \therefore 2^6 = 64$

Hence $P(1970) = (50{,}000)(64) = 3{,}200{,}000$

(b) 1990 $\Rightarrow t = 80 \Rightarrow t/10 = 8 \therefore 2^8 = 512$

Hence $P(1990) = (50{,}000)(512) = 25{,}600{,}000$

(c) 2000 $\Rightarrow t = 90 \Rightarrow t/10 = 9 \therefore 2^9 = 1024$

Hence $P(2000) = (50{,}000)(1024) = 51{,}200{,}000$

31 As we have seen above

(a) $(100{,}000)(4) = 400{,}000$

(b) $(100{,}000)(16) = 1{,}600{,}000$

(c) $(100{,}000)(64) = 6{,}400{,}000$

32 As we have seen above

(a) $(200000)(64) = 12{,}800{,}000$

(b) $(200000)(32) = 6{,}400{,}000$

33 (a) $(25000)(27) = 675000$

(b) $(25000)(81) = 2025000$

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Page 25 Exercises

1 If a is even and b is even, then $a+b$ is even.

From assumptions, $a = 2k$, $b = 2\lambda$.

Then $a+b = 2k + 2\lambda$

$= 2(k+\lambda)$

$a+b = 2\phi$

As $a+b$ can be written as multiple of 2 or as an even integer, $a+b$ must be even.

If a is odd and b is even, then $a+b$ is odd.

From assumptions, $a = 2k+1$, $b = 2\lambda$.

Then $a+b = 2k+1+2\lambda$

$a+b = 2\phi + 1$

$\therefore a+b$ must be odd.

If a is even and b is odd, then $a+b$ is odd. It follows from above.

2 If a is even and b is any positive integer (even or odd), then ab is even.

Let $a = 2k$, $b = 2\lambda + 1$.

$ab = (2k)(2\lambda+1)$

$= 4k\lambda + 2k$

$= 2[2k\lambda + 2k]$

$ab = 2\phi \implies ab$ is even

Let $a = 2k$, $b = 2\lambda$

$ab = (2k)(2\lambda)$

$= 4k\lambda$

$ab = 4\phi \implies ab$ is even

Hence, if a is even and b is any positive integer then ab must be even.

3 Let $\alpha = 2k$.

Then $\alpha^3 = 8k^3 = 2(4k^3) = 2\phi$

Hence α^3 must be even.

4 Let $\alpha = 2k+1$

Then $\alpha^3 = (2k+1)^3 = 8k^3 + 1 + 6k(2k+1)$

$= 8k^3 + 1 + 12k^2 + 6k$

$= 2[4k^3 + 6k^2 + 3k] + 1$

$\alpha^3 = 2\phi + 1$

Hence α^3 must be odd.

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5 Let $n = 2k$. Remember that $(-1)(-1) = 1$.

Then $(-1)^n = (-1)^{2k} = [(-1)(-1)]^k = 1^k = 1$.

Hence, $(-1)^n = 1$ if n is even.

6 Let $n = 2k+1$.

Then $(-1)^n = (-1)^{2k+1} = 1^k(-1) = (1)(-1) = -1$

Hence, $(-1)^n = -1$ if n is odd.

7 Let $m = 2k+1$, $n = 2\lambda+1$.

Then $mn = (2k+1)(2\lambda+1) = 4k\lambda + 2k + 2\lambda + 1 = 2\phi + 1$

Hence mn is odd when m, n are odd.

8 2^4

9 2^3

10 2^5

11 2^2

12 2

13 2^6

14 2^2

15 2^2

16 3

17 3^3

18 3^2

19 3^2

20 3

21 3^0

22 3

23 3^2

24 Let $a - b = 5k$, $x - y = 5\phi$

Then $a + x - b - y = 5k - 5\phi = 5\lambda$

Hence $a + x \equiv b + y \pmod 5$.

Then $ax - by$

$= \alpha x - \alpha y + \alpha y - by$

$= \alpha(x-y) + y(\alpha - b)$

$= \alpha 5\phi + y5k$

$= 5\lambda + 5t$

$= 5\gamma$

Hence $ax \equiv by \pmod 5$.

[Remember a, b, x, y are integers that possess the property $a \equiv b \pmod 5$, $x \equiv y \pmod 5$]

25 The answer follows from question 24. replacing 5 with d.

~~26 Let positive integer be P.~~

(i) $P = 3k$

Then $P^2 = 3\phi$

LHS $= P^2 = (3k)^2 = 9k^2 = 3k$

RHS $= 3\phi$

~~LHS~~. LHS $\equiv$ RHS mod 3.

Hence $P^2 \equiv P \pmod 3$.

(ii) $P = 3k+1$

Then $P^2 = (3k+1)^2 = 9k^2 + 1 + 6k = 3\phi + 1$

Then $P^2 \equiv P \pmod 3$.

(iii) $P = 3k+2$

Then $P^2 = (3k+2)^2 = 9k^2 + 4 + 12k = 3\phi + 4$

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$\in$ → belongs to

26 Let P be a positive integer. Given that P is either of the form $3k$, $3k+1$ and $3k+2$ where k is an integer.

The claim is if P^2 divisible by 3 then P is divisible by 3.

$[\ 3 \mid P^2 \Longrightarrow 3 \mid P\ ]$

(i) $P = 3k$

$P^2 = 9k^2 = 3\phi$

Here $\frac{P^2}{3} \in I$ and $\frac{P}{3} \in I$. So claim holds

(ii) $P = 3k+1$

$P^2 = 9k^2 + 6k + 1 = 3\phi + 1$

Here $\frac{P^2}{3} \notin I$ and $\frac{P}{3} \notin I$. So claim holds

(iii) $P = 3k+2$

$P^2 = 9k^2 + 12k + 4$

Here $\frac{P^2}{3} \notin I$ and $\frac{P}{3} \notin I$. So claim holds.

Hence, $3 \mid P \Longleftrightarrow 3 \mid P^2$ (read as if P square is divisible by 3 if and only if P is divisible by 3 meaning when P square is divisible by 3 so is P.

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Page 39 Exercises

1 (a) $2\alpha = \frac{3}{4}$ (b) $\frac{3\alpha}{5} = -7$ (c) $\frac{-5\alpha}{2} = \frac{3}{8}$

$\alpha = \frac{3}{8}$ $\alpha = \frac{-35}{3}$ $\alpha = -\frac{3}{20}$

2 (a) $3x - 5 = 0$ (b) $-2x + 6 = 1$ (c) $-7x = 2$

$x = \frac{5}{3}$ $x = \frac{5}{2}$ $x = -\frac{2}{7}$

3 (a) $\frac{10}{25} = \frac{2}{5}$ (b) $\frac{3}{9} = \frac{1}{3}$ (c) $\frac{30}{25} = \frac{6}{5}$ (d) $\frac{50}{15} = \frac{10}{3}$ (e) $\frac{45}{9} = \frac{5}{1}$

(f) $\frac{62}{4} = \frac{31}{2}$ (g) $\frac{23}{46} = \frac{1}{2}$ (h) $\frac{16}{40} = \frac{2}{5}$

4 We have $\alpha = \frac{m}{n}$.

From given condition $ab = 1$.

Then $\frac{m}{n} \cdot b = 1 \iff b = \frac{n}{m}$

[So there is a rational number b whose value is exactly $\frac{n}{m}$ such that $\alpha b = b\alpha = 1$. It is the multiplicative inverse]

5 (a) $2x - 7 = 21$

$x = 14$

(b) $3(2x - 5) = 7$

$6x = 22$

$x = \frac{11}{3}$

(c) $(4x - 1)2 = \frac{1}{4}$

$8x - 2 = \frac{1}{4}$

$32x - 8 = 1$

$32x = 9$

$x = \frac{9}{32}$

(d) $-4x + 3 = 5x$

$-9x = 3$

$x = -\frac{1}{3}$

(e) $3x - 2 = -5x + 8$

$8x = 10$

$x = \frac{5}{4}$

(f) $3x + 2 = -3x + 4$

$6x = 2$

$x = \frac{1}{3}$

(g) $\frac{4x}{3} + 1 = 3x$

$4x + 3 = 9x$

$5x = 3$

$x = \frac{3}{5}$

(h) $-\frac{3x}{2} + \frac{4}{3} = 5x$

$\frac{-9x + 8}{6} = 5x$

$-9x + 8 = 30x$

$39x = 8$

$x = \frac{8}{39}$

(i) $\frac{2x - 1}{3} + 4x = 10$

$2x - 1 + 12x = 30$

$14x = 31$

$x = 31/14$

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6 (a) $2x - \frac{3}{7} = \frac{x}{5} + 1$

$5(14x - 3) = (x + 5)7$

$70x - 15 = 7x + 35$

$63x = 50$

$x = \frac{50}{63}$

(b) $\frac{3}{4}x + 5 = -7x$

$3x + 20 = -28x$

$31x = -20$

$x = \frac{-20}{31}$

(c) $\frac{-2}{13}x = 3x - 1$

$-2x = 39x - 13$

$41x = 13$

$x = \frac{13}{41}$

(d) $\frac{4x}{3} + \frac{3}{4} = 2x - 5$

$16x + 9 = 12(2x - 5)$

$16x + 9 = 24x - 60$

$69 = 8x$

$x = \frac{69}{8}$

(e) $\frac{4(1-3x)}{7} = 2x - 1$

$4 - 12x = 14x - 1$

$5 = 26x$

$x = \frac{5}{26}$

(f) $\frac{2-x}{3} = \frac{7}{8}x$

$16 - 8x = 21x$

$29x = 16$

$x = \frac{16}{29}$

7 (a) $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$

$6! = 6 \cdot 120 = 720$

$7! = 7 \cdot 720 = 5040$

$8! = 8 \cdot 5040 = 40320$

$\left[\binom{m}{n} = \frac{m!}{n!(m-n)!} = \frac{m \cdot m-1 \cdot m-2 \cdot \ldots \cdot (m-n)!}{n!\,(m-n)!} = \frac{m \cdot m-1 \ldots m-n+1}{n!} \right]$

(b) $\binom{3}{0} = \frac{3!}{0!\,3!} = 1$ $\binom{3}{3} = \frac{3!}{3!\,0!} = 1$

$\binom{3}{1} = \frac{3!}{1!\,2!} = 3$

$\binom{3}{2} = \frac{3!}{2!\,1!} = 3$

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$\binom{4}{0} = \frac{4!}{0!\,4!} = 1$ $\quad \binom{5}{0} = \frac{5!}{0!\,5!} = 1$ $\quad \binom{5}{5} = \frac{5!}{5!\,0!} = 1$

$\binom{4}{1} = \frac{4!}{1!\,3!} = 4$ $\quad \binom{5}{1} = \frac{5!}{1!\,4!} = 5$

$\binom{4}{2} = \frac{4!}{2!\,2!} = 6$ $\quad \binom{5}{2} = \frac{5!}{2!\,3!} = 10$

$\binom{4}{3} = \frac{4!}{3!\,1!} = 4$ $\quad \binom{5}{3} = \frac{5!}{3!\,2!} = 10$

$\binom{4}{4} = \frac{4!}{4!\,0!} = 1$ $\quad \binom{5}{4} = \frac{5!}{4!\,1!} = 5$

(c) $\binom{m}{n} = \binom{m}{m-n}$

$LHS = \frac{m!}{n!\,(m-n)!} = \frac{m!}{(m-n)!\,(m-(m-n))!} = \binom{m}{m-n} = RHS$

(d) $\binom{m}{n} + \binom{m}{n-1} = \binom{m+1}{n}$

$LHS = \frac{m!}{n!\,(m-n)!} + \frac{m!}{(n-1)!\,(m-n+1)!}$

$= \frac{m!}{n\cdot(n-1)!\cdot(m-n)!} + \frac{m!}{(n-1)!\,(m-n+1)\cdot(m-n)!}$

$= \frac{m!}{(n-1)!\,(m-n)!}\left[\frac{1}{n} + \frac{1}{m-n+1}\right]$

$= \frac{m!}{(n-1)!\,(m-n)!}\left[\frac{m+1}{n(m-n+1)}\right]$

$= \frac{(m+1)!}{n!\,(m+1-n)!}$

$= \binom{m+1}{n} = RHS$

8 Let $a = \frac{m}{n}$ where m, n have no common divisor, $n \neq 0$, $m \neq 0$.

By condition, $\alpha^3 = 2 \iff m^3 = 2n^3 = 2k$. This shows m^3 is even which means m is even.

Also, $2n^3 = 2k$ ➡

8 Let be a a positive rational number, we can write $\alpha = \frac{m}{n}$. Also $a^3 = 2$.

Hence $m^3 = 2n^3$. So $m = 2k$.

So $(2k)^3 = 2n^3$. So $n = 2\phi$.

Hence m, n must have 2 as a common divisor. But m and n have no common divisor. Hence our assumption that a is a rational number must be false.

9 Similarly we can show that there is no positive rational number a such that $\alpha^4 = 2$.

10 Let a be a positive rational number so $a = \frac{m}{n}$.

So $m^2 = 3n^2$. - From problem 26 on page 26 [1, §4], that square of a integer divisible by 3 then so is the integer. Hence, $m = 3k$.

Thus $n = 3\phi$.

By similar argument a cannot be rational.

If $\sqrt{2}$ is rational then its square must be rational too.

$a = \sqrt{2} \iff \alpha^2 = 2$

We know from previous theorem 4 that this is not the case.

11 (a) 1 gives the square 1 and 2 gives the square 4

So the number whose square gives 2 must be between 1 and 2

1.4 gives the square 1.96 and 1.5 gives the square 2.25

So the number whose square gives 2 must between 1.4 and 1.5

Repeating gives us 1.414.

(b) We find the number to be 1.4142.

12 (a), (b) We get 1.732.

13 (a), (b) we get 2.236.

14 (a), (b) 1.259.

15 (a), (b) 1.442.

16 We start with 6 grams. It halves every 3 minutes. Using the previous logic used in exercises of [1, § 3],

$W(t) = 6\left(\frac{1}{2}\right)^{t/3}$ where t is in minutes and $W(t)$ in grams.

(a) $W(3) = \frac{6}{2} = 3g$

(b) $W(27) = 6\left(\frac{1}{2}\right)^{9} \sim 0.0117\,g$

(c) $W(36) = 6\left(\frac{1}{2}\right)^{12} \sim 0.00146\,g$

[You may use fractions for exact answer]

17, 18, 19, 20, 21 can be solved in a similar way.

Page 48 Exercices

1 a) $\frac{2x-1}{3x+2} = 7$

$2x - 1 = 21x + 14$

$19x = -15$

$x = \frac{-15}{19}$

b) $\frac{2-4x}{x+1} = \frac{3}{4}$

$8 - 16x = 3x + 3$

$5 = 19x$

$x = \frac{5}{19}$

c) $\frac{x}{x+5} = \frac{5}{7}$

$7x = 5x + 25$

$2x = 25$

$x = \frac{25}{2}$

d) $2x + 5 = \frac{3x-2}{7}$

$14x + 35 = 3x - 2$

$11x = -37$

$x = \frac{-37}{11}$

e) $\frac{1-2x}{3x+4} = -3$

$2x - 1 = 9x + 12$

$-13 = 7x$

$x = \frac{-13}{7}$

f) $\frac{-2-5x}{-3x-4} = \frac{4}{-3}$

$6 + 15x = -(12x + 16)$

$27x = -22$

$x = \frac{-22}{27}$

g) $\frac{-2-7x}{4} + 1 = \frac{1-x}{5}$

$\frac{-2-7x+4}{4} = \frac{1-x}{5}$

$\frac{2-7x}{4} = \frac{1-x}{5}$

$5(7x-2) = (x-1)4$

$35x - 10 = 4x - 4$

$31x = 6$

$x = \frac{6}{31}$

h) $\frac{3x+1}{4-2x} + \frac{7}{3} = 0$

$\frac{3x+1}{2x-4} = \frac{7}{3}$

$9x + 3 = 14x - 28$

$5x = 31$

$x = \frac{31}{5}$

i) $\frac{-2-4x}{3} = \frac{x-1}{4} + 5$

$\frac{-2-4x}{3} = \frac{x+19}{4}$

$-8 - 16x = 3x + 57$

$19x = -65$

$x = -65/19$

2 a) $\frac{1}{x+y} - \frac{1}{x-y}$

$= \frac{(x-y)-(x+y)}{(x+y)(x-y)}$

$= \frac{x-y-x-y}{x^2-xy+xy-y^2}$

$= \frac{-2y}{x^2-y^2}$

$= RHS$

b) $\frac{x^3-1}{x-1} = 1+x+x^2$

$LHS = (1+x+x^2)(x-1)$

$= x-1+x^2-x+x^3-x^2$

$= x^3-1$

$= RHS$

c) $\frac{x^4-1}{x-1} = 1+x+x^2+x^3$

$LHS = (1+x+x^2+x^3)(x-1)$

$= x-1+x^2-x+x^3-x^2+x^4-x^3$

$= x^4-1$

$= RHS$

d) $\frac{x^n-1}{x-1} = x^{n-1}+x^{n-2}+x^{n-3}+\ldots+x+1$

$LHS = x^n-1$

$= (x^n-x^{n-1})+(x^{n-1}-x^{n-2})+(x^{n-2}-x^{n-3})+(x^{n-3}-x^{n-4})+\ldots+(x^2-x)+(x-1)$

$= x^{n-1}(x-1)+x^{n-2}(x-1)+x^{n-3}(x-1)+\ldots x(x-1)+(x-1)$

$= (x-1)(1+x+x^2+x^3+\ldots+x^{n-2}+x^{n-1})$

$= RHS$

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3 a) $\frac{1}{2x+y} + \frac{1}{2x-y} = \frac{4x}{4x^2-y^2}$

$LHS = \frac{2x-y+2x+y}{(2x+y)(2x-y)}$

$= \frac{4x}{4x^2-y^2}$

$= RHS$

b) $\frac{2x}{x+5} - \frac{3x+1}{2x+1} = \frac{x^2-14x-5}{2x^2+11x+5}$

$LHS = \frac{2x(2x+1)-(3x+1)(x+5)}{(x+5)(2x+1)}$

$= \frac{4x^2+2x-3x^2-15x-x-5}{2x^2+x+10x+5}$

$= \frac{x^2-14x-5}{2x^2+11x+5}$

$= RHS$

c) $LHS = \frac{1}{x+3y} + \frac{1}{x-3y}$

$= \frac{x-3y+x+3y}{(x+3y)(x-3y)}$

$= \frac{2x}{x^2-9y^2}$

$= RHS$

d) $LHS = \frac{1}{3x-2y} + \frac{x}{x+y}$

$= \frac{x+y+3x^2-2xy}{(3x-2y)(x+y)}$

$= \frac{3x^2-2xy+x+y}{3x^2+3xy-2xy-2y^2}$

$= \frac{3x^2-2xy+x+y}{3x^2+xy-2y^2}$

$= RHS$

4 a) $\frac{x^3-y^3}{x-y} = x^2+xy+y^2$

$LHS = (x-y)(x^2+xy+y^2)$

$= x^3+x^2y+xy^2-x^2y-xy^2-y^3$

$= x^3-y^3$

$= RHS$

b) $LHS = (x-y)(x^3+x^2y+xy^2+y^3)$

$= x^4+x^3y+x^2y^2+xy^3$
$-yx^3-x^2y^2-xy^3-y^4$

$= x^4-y^4$

$= RHS$

If we have $x^n-y^n=0$.

We $x=y$ satisfies this equation

So $(x-y)$ is a factor.

$x^n-y^n = (x^n-x^{n-1}y)+(x^{n-1}y-x^{n-2}y^2)+x^{n-2}y^2+\ldots-x^2y^{n-2}+(x^2y^{n-2}$
$-xy^{n-1})+(xy^{n-1}-y^n)$

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Consider the pairs

$$x^n - x^{n-1}y = (x-y)x^{n-1}$$
$$x^{n-1}y - x^{n-2}y^2 = (x-y)x^{n-2}y$$
$$\vdots$$
$$xy^{n-1} - y^n = y^{n-1}(x-y)$$

So finally

$$x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + x^2y^{n-3} + xy^{n-2} + y^{n-1})$$

Hence

$$\frac{x^n - y^n}{x-y} = x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + x^2y^{n-3} + xy^{n-2} + y^{n-1}$$

c) $$LHS = \left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2$$

$$= \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2}$$

$$= \frac{1 - 2t^2 + t^4 + 4t^2}{1 + 2t^2 + t^4}$$

$$= \frac{1 + 2t^2 + t^4}{1 + 2t^2 + t^4}$$

$$= 1$$

$$= RHS$$

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5 a) LHS $= (x+1)(x^2-x+1)$

$= x^3 - x^2 + x + x^2 - x + 1$

$= x^3 + 1$

$=$ RHS

b) LHS $= (x+1)(x^4 - x^3 + x^2 - x + 1)$

$= x^5 - x^4 + x^3 - x^2 + x$
$+ x^4 - x^3 + x^2 - x + 1$

$= x^5 + 1$

$=$ RHS

c) LHS $= (x+1)(x^{n-1} - x^{n-2} + x^{n-3} - \ldots - x + 1)$

$= x^n - x^{n-1} + x^{n-2} - \ldots - x^2 + x$
$+ x^{n-1} - x^{n-2} + x^{n-3} - \ldots - x + 1$

$= x^n + 1$

$=$ RHS

6 By definition of speed = distance / time, we can say the fodlowing. Let x be time required.

$2/3 : 1 :: 5/4 : x$

$\frac{2}{3} = \frac{5}{4x}$

$8x = 15$

$x = \frac{15}{8}$ seconds

7 By defintion, $d = \frac{m}{v}$.

a) $\frac{3/10}{2/3} = \frac{9}{20}$ lb / in^3 b) $\frac{6}{4/3} = \frac{9}{2}$ lb/in^3 c) $v = \frac{m}{d}$; $\frac{15}{2/3} = \frac{45}{2}$ in^3

8, 9, 10 can be solved by plugging in the numbers.

11 We have 4.2 in^3 of alchod and 7.8 in^3 of water aworing to question

a) If alchod becomes 20%, water is 80%. Also 20% of alchod is 4.2 in^3
So our final solution's total volume is $\frac{4.2}{0.2} = 21$ in^3.
So our amount of water is 80% of 21 in^3 = 16.8 in^3
So finally we need to add 9 in^3 of water.

(or)

As water is the only thing added we get 21 in^3 − 12 in^3 = 9 in^3. Hence water added needs to be exactly 9 in^3.

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11 b, c can be solved in a similar fashion.

12 Consider the wind to be unfavorable for t and favourable for $4-t$ hours.

Then by definition of speed we can say,

$$900(4-t) + 500t = 3000$$

$$9(4-t) + 5t = 30$$

$$36 - 9t + 5t = 30$$

$$6 = 4t$$

$$t = \frac{3}{2} \text{ hours}$$

So the wind was favourable for $\frac{5}{2}$ hours. [Should have set t as favourable time for direct result but oh well]

13, 14, 15, 16, 17, 18 can be solved similar way.

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To solve all questions in one shot we can show the general case

$$ax + by = u$$
$$cx + dy = v$$

Solving for x,

$$adx + bdy = du$$
$$bcx + bdy = bv$$

$$adx + bdy - bcx - bdy = du - bv$$
$$x[ad - bc] = du - bv$$
$$\boxed{x = \frac{du - bv}{ad - bc}}$$

Solving for y,

$$acx + bcy = cu$$
$$acx + ady = av$$

$$acx + bcy - acx - ady = cu - av$$
$$y(ad - bc) = av - cu$$
$$\boxed{y = \frac{av - cu}{ad - bc}}$$

For verification,

Put x, y value in equation 1, LHS =

$$a\left[\frac{du - bv}{ad - bc}\right] + b\left[\frac{av - cu}{ad - bc}\right]$$

$$= \frac{adu - abv + abv - bcu}{ad - bc}$$

$$= \frac{u(ad - bc)}{ad - bc}$$

$$= u = RHS$$

Put x, y value in equation 2, LHS =

$$c\left[\frac{du - bv}{ad - bc}\right] + d\left[\frac{av - cu}{ad - bc}\right]$$

$$= \frac{cdu - bcv + adv - cdu}{ad - bc}$$

$$= \frac{v(ab - bc)}{ad - bc}$$

$$= v = RHS$$

You can proceed with questions 1 through 11.

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Page 59 Exercises

Once again we can look at the general case of 3 equations and 3 unknowns.

$$\begin{aligned} ax + by + cz &= u \\ dx + ey + fz &= v \\ gx + hy + iz &= w \end{aligned}$$

In matrix form $AX = B$:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

$$D = \det(A) = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

By Cramer's rule,

$$x = \frac{D_x}{D} = \frac{u(ei - fh) - b(vi - fw) + c(vh - ew)}{D}$$

$$y = \frac{D_y}{D} = \frac{a(vi - fw) - u(di - fg) + c(dw - vg)}{D}$$

$$z = \frac{D_z}{D} = \frac{a(ew - vh) - b(dw - vg) + u(dh - eg)}{D}$$

If $D = 0$, the simultaneous equations have no solution or infinitely many solutions. Do it by the method of elimination or use elementary row operations to find a contradiction.

Page 63 Exercises

For this section of exercises it is beneficial to define the objects E and I in terms of numbers so that we can perform operations and see what properties these objects follow. Let $n \in \mathbb{Z}$. Then we define E and I by

$$E(n) = 2n, \qquad I(n) = 2n + 1.$$

Examples of such systems of two objects are $\{2, 3\}$, $\{4, 17\}$, or $\{-8, -21\}$.

1. a) Let $E = 2h$, $E' = 2t$ and $I = 2k + 1$, where $h, t, k \in \mathbb{Z}$.
Commutativity.

$$\text{LHS} = E + I = 2h + 2k + 1 = 2m + 1,$$
$$\text{RHS} = I + E = 2k + 1 + 2h = 2m + 1,$$

where $m = h + k$.
Associativity.

$$\text{LHS} = E + (I + E') = 2h + (2k + 1 + 2t) = 2v + 1,$$
$$\text{RHS} = (E + I) + E' = (2h + 2k + 1) + 2t = 2v + 1,$$

where $v = h + k + t$.
Hence addition of E and I is commutative and associative because LHS and RHS are equal quantities.

Zero element. If an object is to be the zero element for addition in this system (which consists of two objects, an even object E and an odd object I), then adding it to an even object or an odd object must not change whether that object is even or odd. That is, if X is even then $X +$ zero element must be even, and if X is odd then $X +$ zero element must be odd. Remember we are dealing with a new system $\{E, I\}$, not numbers like 1, $\frac{1}{7}$ or -3. We now show that E plays the role of the zero element for addition in this system. Let $X = 2w$ if X is even and $X = 2s + 1$ if X is odd.

$$\begin{aligned} X \text{ even:} \quad & E + X = 2h + 2w = 2(h + w) = 2d, && \text{so } X \text{ stays even,} \\ X \text{ odd:} \quad & E + X = 2h + 2s + 1 = 2(h + s) + 1 = 2f + 1, && \text{so } X \text{ stays odd.} \end{aligned}$$

Additive inverse. The additive inverse of I can be found from the definition of the additive inverse (a number summed with its additive inverse gives zero) and the property $a + b = 0 \iff b = -a$. Let $I(h) = 2h + 1$ and let I' denote the additive inverse of I as numbers. Then

$$I(h) + I' = 0 \iff I' = -I(h) = -2h - 1 = 2(-h - 1) + 1 = I(h'), \quad h' = -h - 1.$$

So I' is itself an odd object. Inside the system $\{E, I\}$, adding I and I' gives

$$I + I' = (2h + 1) + (2h' + 1) = 2(h + h' + 1) = 2g,$$

which is even, i.e. the zero element E. Hence the additive inverse of I in this system is I itself ($I + I = E$). Similarly E is its own additive inverse, since $E + E' = 2(h + t)$ is even.

b) Let $E = 2h$, $E' = 2t$ and $I = 2k + 1$, where $h, t, k \in \mathbb{Z}$.
Commutativity.

$$\text{LHS} = EI = (2h)(2k + 1) = 4hk + 2h = 2m,$$
$$\text{RHS} = IE = (2k + 1)(2h) = 4hk + 2h = 2m,$$

where $m = 2hk + h$.
Associativity.

$$\text{LHS} = E(IE') = (2h)\big((2k + 1)(2t)\big) = 8hkt + 4ht = 2v,$$
$$\text{RHS} = (EI)E' = \big((2h)(2k + 1)\big)(2t) = 8hkt + 4ht = 2v,$$

where $v = 4hkt + 2ht$.
Hence multiplication of E and I is commutative and associative because LHS and RHS are equal quantities.

Identity and zero elements. Let $I' = 2j+1$ be any odd object, and let $X = 2w$ (even) or $X = 2s+1$ (odd).

$$\begin{aligned} I'X &= (2j+1)(2w) = 2\big(w(2j+1)\big) && \text{even, so } X \text{ stays even,} \\ I'X &= (2j+1)(2s+1) = 2(2js+j+s)+1 && \text{odd, so } X \text{ stays odd.} \end{aligned}$$

Multiplying by an odd object does not change whether X is even or odd, so I plays the role of the one element for multiplication in this system. On the other hand,

$$\begin{aligned} EX &= (2h)(2w) = 2(2hw) && \text{even,} \\ EX &= (2h)(2s+1) = 2\big(h(2s+1)\big) && \text{even,} \end{aligned}$$

so multiplying by E always gives an even object, whatever X is. Hence E plays the role of the zero element for multiplication in this system.

Distributivity. Multiplication is distributive over addition, i.e. $E(E'+I) = EE' + EI$:

$$\begin{aligned} \text{LHS} &= (2h)(2t+2k+1) = 4ht+4hk+2h = 2q, \\ \text{RHS} &= (2h)(2t)+(2h)(2k+1) = 4ht+4hk+2h = 2q, \end{aligned}$$

where $q = 2ht+2hk+h$.
Hence multiplication is distributive for E and I because LHS and RHS are equal quantities.

Page 68 Exercises

1. a) **Claim.** If $a \in \mathbb{R} \implies a^2 > 0$. Here a may be positive, negative or equal to zero. If a is positive then by POS 1 a^2 must be positive. If a is zero then a^2 is positive (we define 0 to be positive in the textbook but it is definitely debatable). If a is negative than $-a$ is positive then by POS 1 a^2 is positive.

   b) If $a \geq 0, b < 0 \implies ab < 0$. By POS 2 and given information, a and $-b$ are positive. Hence by POS 2 $-(ab)$ must be positive. Then by definition of the minus sign ab must be negative.

   c) If $a, b < 0 \implies ab \geq 0$. By POS 2, $-a, -b$ must be positive then by POS 1 ab must be positive.

2. If $a > 0 \implies a^{-1} > 0$. We know $a \cdot a^{-1} = 1$. We know 1 is positive. So both a and a^{-1} must be both positive or negative. Because $a > 0$ it must mean $a^{-1} > 0$. If not than we find that -1 is positive which is absurd.

3. If $a < 0 \implies a^{-1} < 0$. Follow from above.

4. If $a, b > 0 \implies \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$. By POS 1, $\frac{a}{b} > 0 \implies \sqrt{\frac{a}{b}} > 0$. By definition of the square root $a, b > 0 \implies \sqrt{a}, \sqrt{b} > 0$. Also $\left(\sqrt{\frac{a}{b}}\right)^2 = \left(\frac{\sqrt{a}}{\sqrt{b}}\right)^2 = \frac{a}{b}$. The equality holds both in value and parity. Hence the statement must be true.

5. $\frac{1}{1-\sqrt{2}} = \frac{1+\sqrt{2}}{1-2} = -(1+\sqrt{2})$.

6. $\frac{1}{1+\sqrt{2}} = 2 - \sqrt{3}$.

7. $\frac{1}{3+\sqrt{5}} = \frac{3}{4} - \frac{1}{4}\sqrt{5}$.

8. $\frac{1}{a+b\sqrt{2}} = \frac{a}{a^2-2b^2} + \frac{b}{2b^2-a^2}\sqrt{2}$.

9. $\frac{1}{a+b\sqrt{3}} = \frac{a}{a^2-2b^2} + \frac{b}{2b^2-a^2}\sqrt{3}$.

10. $(x + y\sqrt{5})(z + w\sqrt{5}) = (xz + 5yw) + (xw + yz)\sqrt{5}$.

11. $(x + y\sqrt{a})(z + w\sqrt{a}) = (xz + ayw) + (xw + yz)\sqrt{a}$.

12. a) $\dfrac{\sqrt{2x+3}+1}{4} = \dfrac{x+1}{2\left(\sqrt{2x+3}-1\right)}$

    b) $\dfrac{\sqrt{1+x}-3}{2} = \dfrac{x-8}{2\left(\sqrt{1+x}+3\right)}$

    c) $\dfrac{\sqrt{x-h}-\sqrt{x}}{h} = \dfrac{-1}{\sqrt{x-h}+\sqrt{x}}$

    d) $\dfrac{\sqrt{x-h}+\sqrt{x}}{h} = \dfrac{-1}{\sqrt{x-h}-\sqrt{x}}$

    e) $\dfrac{\sqrt{x+h}+\sqrt{x}}{h} = \dfrac{1}{\sqrt{x+h}-\sqrt{x}}$

    f) $\dfrac{\sqrt{x+2h}-\sqrt{x}}{h} = \dfrac{2}{\sqrt{x+2h}+\sqrt{x}}$

13. a) $|x-1| = 2 \implies x - 1 = \pm 2 \implies x = 3$ or $x = -1$.

    b) $|x| = 5 \implies x = 5$ or $x = -5$.

    c) $|x-3| = 4 \implies x - 3 = \pm 4 \implies x = 7$ or $x = -1$.

    d) $|x+1| = 6 \implies x + 1 = \pm 6 \implies x = 5$ or $x = -7$.

    e) $|x+4| = 3 \implies x + 4 = \pm 3 \implies x = -1$ or $x = -7$.

    f) $|x-2| = 1 \implies x - 2 = \pm 1 \implies x = 3$ or $x = 1$.

14. a) $|2x-1| = 3 \implies 2x - 1 = \pm 3 \implies x = 2$ or $x = -1$.

    b) $|3x+1| = 2 \implies 3x + 1 = \pm 2 \implies x = \frac{1}{3}$ or $x = -1$.

    c) $|2x+1| = 4 \implies 2x + 1 = \pm 4 \implies x = \frac{3}{2}$ or $x = -\frac{5}{2}$.

    d) $|3x-1| = 1 \implies 3x - 1 = \pm 1 \implies x = \frac{2}{3}$ or $x = 0$.

    e) $|4x-5| = 6 \implies 4x - 5 = \pm 6 \implies x = \frac{11}{4}$ or $x = -\frac{1}{4}$.

15. a) $\dfrac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}} = \dfrac{x-y}{\left(\sqrt{x}-\sqrt{y}\right)^2}$

b) $\dfrac{\sqrt{x+y}-\sqrt{y}}{\sqrt{x+y}+\sqrt{y}} = \dfrac{x}{\left(\sqrt{x+y}+\sqrt{y}\right)^2}$

c) $\dfrac{\sqrt{x+1}+\sqrt{x-1}}{\sqrt{x+1}-\sqrt{x-1}} = \dfrac{2}{\left(\sqrt{x+1}-\sqrt{x-1}\right)^2}$

d) $\dfrac{\sqrt{x-3}+\sqrt{x}}{\sqrt{x-3}-\sqrt{x}} = \dfrac{-3}{\left(\sqrt{x-3}-\sqrt{x}\right)^2}$

e) $\dfrac{\sqrt{x+y}-1}{3+\sqrt{x+y}} = \dfrac{x+y-1}{(3+\sqrt{x+y})\left(\sqrt{x+y}+1\right)}$

f) $\dfrac{\sqrt{x+y}+x}{\sqrt{x+y}} = \dfrac{x+y-x^2}{\sqrt{x+y}\left(\sqrt{x+y}-x\right)}$

16. a) $\dfrac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}} = \dfrac{\left(\sqrt{x}+\sqrt{y}\right)^2}{x-y}$

b) $\dfrac{\sqrt{x+y}-\sqrt{y}}{\sqrt{x+y}+\sqrt{y}} = \dfrac{\left(\sqrt{x+y}-\sqrt{y}\right)^2}{x}$

c) $\dfrac{\sqrt{x+1}+\sqrt{x-1}}{\sqrt{x+1}-\sqrt{x-1}} = \dfrac{\left(\sqrt{x+1}+\sqrt{x-1}\right)^2}{2}$

d) $\dfrac{\sqrt{x-3}+\sqrt{x}}{\sqrt{x-3}-\sqrt{x}} = -\dfrac{\left(\sqrt{x-3}+\sqrt{x}\right)^2}{3}$

e) $\dfrac{\sqrt{x+y}-1}{3+\sqrt{x+y}} = \dfrac{4\sqrt{x+y}-(x+y)-3}{9-(x+y)}$

f) $\dfrac{\sqrt{x+y}+x}{\sqrt{x+y}} = \dfrac{x+y+x\sqrt{x+y}}{x+y}$

17. $\sqrt{x-1} < 3+\sqrt{x} \therefore \nexists x \in \mathbb{R} \mid \sqrt{x-1} = 3+\sqrt{x}$. In other words the left hand function is always less than the right hand function and the functions do not intersect for all values in the domain of the functions $\sqrt{x-1}$ and $3+\sqrt{x}$.

18. Squaring both sides gives $x-1 = 9-6\sqrt{x}+x$, so $6\sqrt{x} = 10$ and $\sqrt{x} = \frac{5}{3}$. Hence $x = \frac{25}{9}$. Check: $\sqrt{\frac{25}{9}-1} = \frac{4}{3} = 3-\frac{5}{3}$.

19. a) $\sqrt{x-2} = 3+2\sqrt{x}$. Squaring gives $x-2 = 9+12\sqrt{x}+4x$, so $-3x-11 = 12\sqrt{x}$. The left side is negative for $x \geq 0$ and the right side is nonnegative, so **no solution**.

b) $\sqrt{x-2} = 3-2\sqrt{x}$. Squaring gives $x-2 = 9-12\sqrt{x}+4x$, so $3x-12\sqrt{x}+11 = 0$. Let $t = \sqrt{x}$, so $3t^2-12t+11 = 0$ and $t = 2 \pm \frac{\sqrt{3}}{3}$. We need $3-2t \geq 0$, i.e. $t \leq \frac{3}{2}$, so we reject $t = 2+\frac{\sqrt{3}}{3}$. For $t = 2-\frac{\sqrt{3}}{3}$, we have $x \geq 2$ and $3-2t \geq 0$, so it is valid. Hence $x = \left(2-\frac{\sqrt{3}}{3}\right)^2 = \frac{13-4\sqrt{3}}{3}$.

c) $\sqrt{x+3} = 1+\sqrt{x}$. Squaring gives $x+3 = 1+2\sqrt{x}+x$, so $\sqrt{x} = 1$ and $x = 1$. Check: $\sqrt{4} = 2 = 1+1$.

d) $\sqrt{x+3} = 1-\sqrt{x}$. Squaring gives $x+3 = 1-2\sqrt{x}+x$, so $\sqrt{x} = -1$, which is impossible. **No solution**.

e) $\sqrt{x-4} = 3+\sqrt{x}$. Squaring gives $x-4 = 9+6\sqrt{x}+x$, so $6\sqrt{x} = -13$, which is impossible. **No solution**.

f) $\sqrt{x-4} = 3-\sqrt{x}$. Squaring gives $x-4 = 9-6\sqrt{x}+x$, so $\sqrt{x} = \frac{13}{6}$ and $x = \frac{169}{36}$. Check: $\sqrt{\frac{169}{36}-4} = \frac{5}{6} = 3-\frac{13}{6}$.

20. Since $b-a = -(a-b)$, it suffices to show that $|-c| = |c|$ for any number c. Let $c = a-b$.

- If $c \geq 0$, then $-c \leq 0$, so $|-c| = -(-c) = c = |c|$.
- If $c < 0$, then $-c > 0$, so $|-c| = -c = -(-c) = c = |c|$.

In both cases $|-c| = |c|$, hence $|b-a| = |-(a-b)| = |a-b|$.